TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Principles of Loop Formation in Circular, Flat and Warp Knitting Machines

The loop-forming cycle stage by stage: the latch-needle cycle on the circular machine, the carriage-cam cycle of the V-bed flat machine, and the swinging, shogging guide bars of warp knitting — drawn so you can follow them on the machine, inched by hand.

TE-3102 · KNITTING TECHNOLOGY SESSIONAL · STUDY 2 · LEVEL 3 TERM 1

1. Object and the elements every cycle uses

Object: to observe and explain, stage by stage, how one new loop is formed and intermeshed with the previous course on each machine family of Study 1 — and to be able to stop the machine (inched by hand, with the instructor) at any stage and name what needle, sinker and yarn are doing at that instant.

However the machine is built, loop formation needs the same working elements; the practical study is learning where each lives on each machine: **a needle** (holds the old loop, receives new yarn, draws it through); **a yarn-presenting element** (carrier/feeder in weft knitting, guide in warp knitting); **a holding element** (sinkers; in warp knitting also the trick plate) keeping the formed fabric from rising with the needles; **a knocking-over surface** (sinker throat/edge, or trick plate) over which the old loop is cast off; and **take-down** drawing the fabric away under steady tension.

Terms. A **wale** is a column of loops (one needle, one wale); a **course** is a row of loops knitted in one cycle. The **stitch (loop) length** — the yarn in one loop — is the most important quality variable in knitting, fixing fabric weight, density, width and elasticity more than any other setting.

2. The weft knitting cycle on the latch needle (circular machine)

Follow one needle of the circular machine through one feeder (Study 1, Figure 2, shows the cam track), the sinker working in counter-rhythm beside it. There are seven stages; learn them in order, with the sinker’s job at each.

STAGE	NEEDLE AND YARN	SINKER AND FABRIC
1. Rest (knit) position	Needle at its lowest working position; the last-formed loop rests in the hook, latch	Sinker forward; fabric held in the sinker throats under take-down tension.

	closed over it.	
2. Clearing (raising)	The clearing cam lifts the needle. The old loop, held down by the sinker, slides down the stem, pushes the latch open as it passes, and rests on the stem *below* the open latch. The hook is empty and open.	Sinker fully forward, nose over the fabric, preventing it rising with the needle. **Incomplete clearing leaves the old loop in the hook — the classic cause of a dropped stitch or double loop.**
3. Yarn feeding	At the top of the rise the carrier lays the new yarn across the open hook, just above the open latch.	Sinker still forward; the yarn is laid over the sinker throat too, controlling the loop about to be drawn.
4. Latch closing	The stitch cam draws the needle down; the old loop slides up the stem and closes the latch over the hook, trapping the new yarn inside.	Sinker eases back so the old loop can travel up and over the closed hook.
5. Knock-over (cast-off)	The needle descends further; the old loop slips over the closed latch and off the needle head, over the knock-over edge. It now hangs around the new yarn in the hook.	Sinker back; the new loop is supported in the sinker throat.
6. Loop drawing	The needle continues down to the depth set by the stitch cam, pulling the new yarn through the old loop. The length drawn through *is* the stitch length — why the stitch-cam setting (verified by measuring a course, Study 5) governs fabric quality.	Sinker throat receives and sizes the new loop; take-down tension must be steady or the drawn length varies loop to loop.
7. Holding and take-down	The new loop rests in the hook; the needle returns to rest position for the next feeder; the old loop is now intermeshed fabric.	Take-down draws the fabric down by one course spacing, presenting the new row at sinker level for the next cycle.

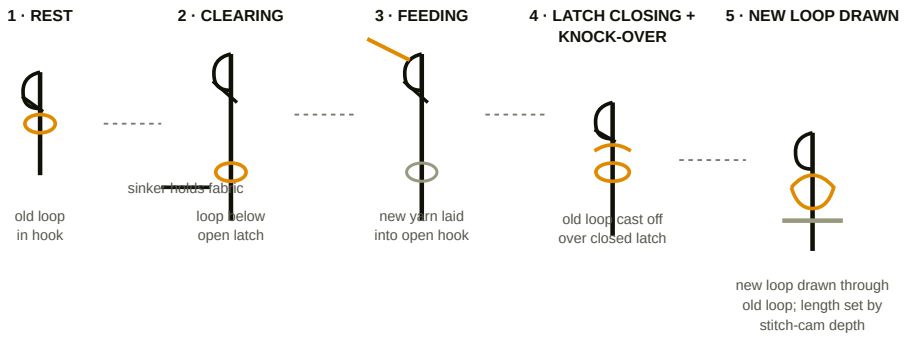


Figure 1 — The latch-needle loop-forming cycle in five drawn positions (stages 1, 2, 3, 4–5 combined, and 6–7 combined). Orange shows yarn: the old loop in the hook, then below the opened latch, the new yarn fed in, the old loop knocked over, and the new loop drawn through it.

WHAT DECIDES A GOOD CYCLE — THREE CHECKS WHILE INCHING THE MACHINE

(1) **Clearing height:** at full rise every old loop must sit below its latch. (2) **Feeding position:** the carrier must lay yarn inside the open hook — yarn laid outside the hook is missed entirely (hole or press-off). (3) **Draw-down depth:** all stitch cams at the same setting — one feeder drawing deeper shows as a horizontal band (barré), its course being heavier than its neighbours.

3. Loop formation on the flat V-bed machine

The flat machine uses the same latch needle and the same seven stages — what changes is *how the needle is driven* and *how the yarn arrives*:

- **The carriage does the cam work.** The needles lie still until the carriage passes; its cam plates lift each butt in turn (clearing), hold it while the yarn carrier — running just ahead on its rail — lays yarn into the open hooks across the working width, then draw the needles down through knock-over and loop drawing. One traverse knits one course, needle by needle in sequence across the bed — the loops of a course are formed one after another, not all at the same stage.
- **Two beds, two directions of loop.** Front-bed needles draw loops towards the front, back-bed needles towards the back; where both beds knit in one course the result is rib, the two sets intermeshing face to face. The inverted V (Figure 2) lets the hooks pass each other at the top without colliding, and lets a loop be **transferred** to the facing needle of the other bed — the basis of shaping and links-links (purl) fabric.
- **Brushes replace part of the sinker's work,** flicking the latches open before feeding; holding-down sinkers or the take-down keep the fabric from rising during clearing.
- **Tuck and miss are selected in the carriage:** a needle raised only part-way (old loop stays in the hook while new yarn is fed) gives a **tuck**; a needle not raised gives a **miss (float)**. Find these cam

positions — they are the flat machine's patterning alphabet.

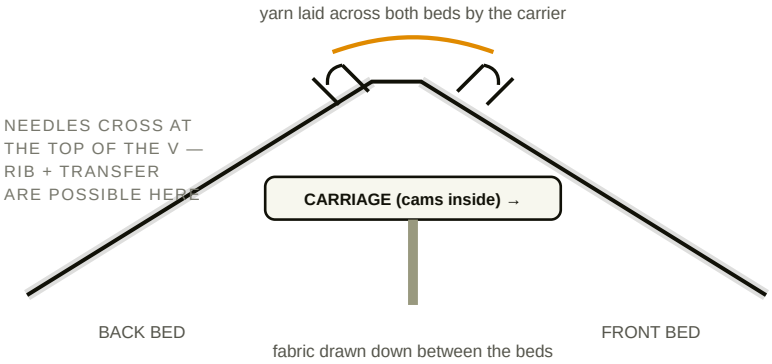


Figure 2 — Cross-section of the V-bed flat machine. The two beds meet at an angle; needles from opposite beds can knit into the same course (rib) or exchange loops (transfer), and the traversing carriage carries the cams that work every needle in sequence.

4. Loop formation in warp knitting

Warp knitting forms loops **along** the fabric, not across it: every needle makes its own wale, fed by a guide that stays with that needle region for the whole beam. The guide bar performs two movements around the needle line, and all warp-knit structure is built from them:

- **Swinging** — the bar swings the guides *through* the needle line, from behind the needles to the front (hook side) and back. The swing lays no loop; it positions the guides for the sideways movements.
- **The overlap** — with the guides at the front, the bar **shogs** (shifts sideways) by one (sometimes two) needle spaces. When the guides swing back, the yarn lies across the hook of the needle moved to, and becomes the **head of the new loop**.
- **The underlap** — with the guides behind the needles, the bar shogs again before the next swing, laying the diagonal yarn behind the fabric that **connects one wale to another**. Without underlaps of at least one space from at least one bar, the wales would not be joined at all — the "fabric" would fall apart into separate chains.

The cycle in order (follow it inched by hand): (1) needle bar rises, old loops sliding below the hooks; (2) guides swing through to the front and shog the **overlap**; (3) guides swing back, leaving yarn on the hooks; (4) behind the needles the bar shogs the **underlap** while the needles begin to fall; (5) needles draw down — sinkers holding the old loops — old loops close the latches (or closing wires shut the hooks) and are knocked over the trick plate as new yarn is drawn through; (6) fabric is taken up by one course spacing, and the cycle repeats several hundred times a minute.

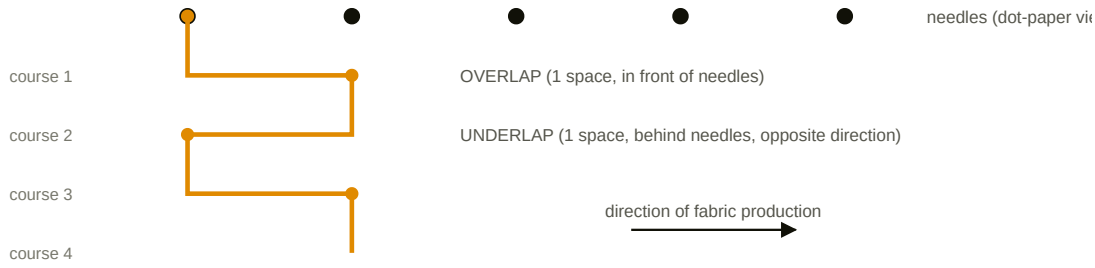


Figure 3 — Lapping diagram of a 1-and-1 tricot lap, drawn on dot paper (dots = needles, one guide's yarn traced in orange). Each course the guide overlaps one needle space in front of the needles and underlaps one space behind them in the opposite direction, so successive courses zig-zag between two wales and the underlaps bind the wales together.

4.1 Reading a lapping movement — what to record

- **Overlap length** — needle spaces shogged in front (usually one; two gives heavier structures).
- **Underlap length and direction** — spaces shogged behind the needles, in the *same* direction as the overlap (**open lap**) or the *opposite* direction (**closed lap**, Figure 3); the two give fabrics of different appearance and stability.
- **Pillar (chain) stitch, the limiting case** — a bar overlapping the *same* needle every course with zero underlap (chain notation 1-0 / 0-1, open lap) makes only a chain in one wale; a second bar with real underlaps must always accompany it.
- **Two bars together** — in standard two-bar tricot both bars make 1-and-1 laps in opposite phases: every wale receives an overlap every course and the underlaps cross behind, giving a light, stable fabric. Record the laboratory machine's bars in this form.

5. The three cycles compared

POINT	WEFT — CIRCULAR	WEFT — FLAT V-BED	WARP
One cycle knits	One course per feeder, around the cylinder	One course per traverse, needle by needle across the bed	One course across *all* needles at once
Yarn laid by	Carrier at each feeder	Carrier running ahead of the carriage	Guides — one per needle — swung through and shogged
Old loop held by	Sinkers	Holding-down sinkers/take-down; brushes open latches	Sinkers and trick plate

What binds loop to loop	Next course drawn through the previous course	Same weft principle, plus transfer between beds	Underlaps behind the needles, joining wales
Stitch length set by	Stitch-cam draw-down depth (and feeder metering)	Stitch-cam values in the carriage, per bed	Run-in per rack from each beam

6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Name the seven stages of the latch-needle cycle in order, and state at which stage the stitch length is actually decided.
2. During clearing, what opens the latch on a circular machine, and what fault follows if a needle under-rises?
3. Define overlap and underlap, and explain why pillar stitch alone falls apart.
4. Figure 3's lap is closed. What single change to the underlap direction makes the lap open, and what happens behind the needles?

Answer key

1. Rest → clearing → feeding → latch closing → knock-over → loop drawing → holding/take-down. Stitch length is decided at **loop drawing**: the stitch cam's draw-down depth fixes the yarn pulled through the old loop.
2. The **old loop itself**: held down by the sinker, it slides down the stem as the needle rises and pushes the latch open before passing below it. If the needle under-rises, the loop never clears below the latch and cannot be cast off cleanly — a doubled loop, or a **dropped stitch** if it slips off the hook.
3. The **overlap** is the guide's shog in front of the needles, laying yarn across a hook to form the new loop's head; the **underlap** is the shog behind the needles, laying the diagonal yarn joining wale to wale. Pillar stitch has no underlaps: each wale is an independent chain, so the chains separate.
4. Shog the underlap in the **same** direction as the overlap: the yarn then progresses across the needles course by course (basis of atlas movements) instead of zig-zagging between the same two wales (Figure 3).